Development of an Adaptive Model to Assess Suitability of Hyperloop Systems Across Diverse Loci

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Introduction

The growing demand for high-speed passenger and freight transportation is placing increasing pressure on existing rail and aviation networks, highlighting the potential of Hyperloop technology**.** While current research has focused on engineering challenges such as propulsion, tube design, and structural feasibility, limited attention has been given to where Hyperloop systems should be built and how passenger and freight services should be allocated.

This research addresses that gap by developing a corridor selection and passenger–freight allocation framework based on measurable socio-economic and geographic factors. The framework evaluates corridor characteristics including population, GDP, travel demand, freight activity, terrain, distance, and regulatory conditions to identify routes with the greatest potential for network value.

This framework was tested using three case studies: West Africa, Southeast Asia, the Middle East. Each case study represents different transportation, economic, and regulatory environments. This framework provides a structured decision-making tool to support future Hyperloop planning and investment by integrating technical feasibility with broader socio-economic considerations.

Background & Literature Review

Economics

In 2026, the Mid-Ohio Regional Planning Commission (MORPC) published a feasibility study for a Chicago-Columbus-Pittsburgh Hyperloop corridor with a 30-year lifespan. The MORPC cited a variety of benefits, including expedited commutes, a diversified transportation profile, 300 billion USD in profits, and reduced CO_2 emissions via 1.9 billion diverted automobile trips.

Transport Canada (2026) and NASA's Glenn Research Center (2016) published separate analyses projecting capital expenses of 17-27 million USD per mile of track, disregarding the cost of land acquisition. This would place passenger fare at roughly 60 USD for a 300-mile voyage, or 0.20 USD per mile. The MORPC suggested that this fare target would result in an annual ridership of over two million passengers.

Governance and Regulation

Contemporary border clearance frameworks for air travel are ill-suited to high-speed rail. Frequent stops across regions of densely-clustered ports of entry, as in the Schengen region, threaten the efficiency underpinning Hyperloop's implementation.

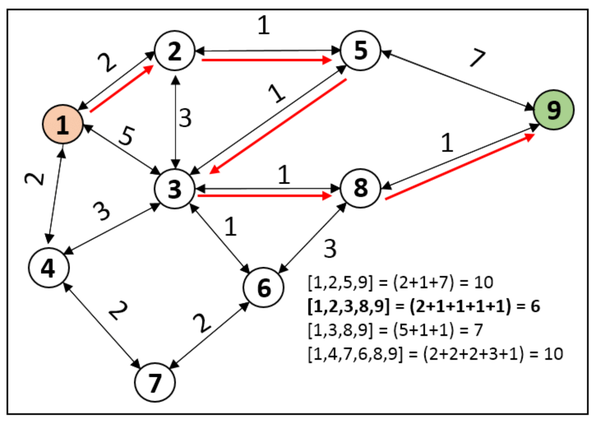


Figure 4 and 5: Visualized example of Dijkstra's algorithm in action, with the shortest path highlighted in red (left). Example of Dijkstra's algorithm in action, with each stage of current node selection explicitly tabulated (right).

Methodology

Dijkstra's Algorithm

Dijkstra's algorithm is a fundamental concept of computing used to solve shortest-path problems in graphical analysis. Characterized as a greedy method – at each decision step (node), the current node is selected from a set of candidates such that the path between the source and current nodes has the lowest "cost." For a network of *N* nodes, the resulting time complexity is $O(N^2)$. Mathematically, the algorithm is represented by Equation 1:

$$d_j = \min\left(d_j, d_i + c_{ij}\right)$$

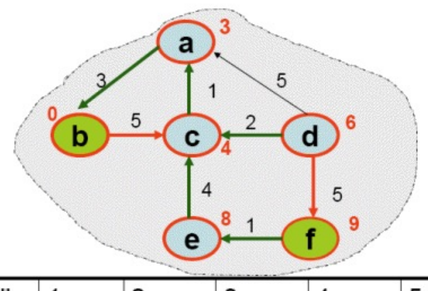
where in d_j is the best-known distance to current node j, d_i is the final distance to source node i, and c_{ij} is the edge weight between nodes i and j. The greedy method works on the assumption that all nodes of the graph are connected, that the edges are directed or undirected, and that the edge weights are non-negative.

Track Allocation

Hyperloop track allocation benefits from a decision-making framework based on Dijkstra's algorithm, minimizing installation costs based on known socio-economic data. Our study treats the sampled Hyperloop networks as directed weighted graphs, wherein each node represents a boarding station or hub, edges are track segments, and edge weights reflect a composite of metrics including travel fares, travel time, switching delays, and the operating constraints inherent to passenger or freight vehicles.

Corridor Selection

The corridor selection tool is implemented as a modular Python pipeline, wherein each stage operates on a shared list of node objects, each tagged with an ID, name, coordinates, and a dictionary of attributes that grows with passage through later stages. The *main.py* program calls each stage in sequence and transposes raw geographic data into a network graph by querying from OpenStreetMap and the Open-Elevation API using the OSMnx library. Candidate nodes are passed through an enrichment stage wherein each is scored using the four attributes.



vertex	initially	1	2	3	4	5	6
a	∞	3b √	3b √	3b √	3b √	3b √	3b
b	0 -	0- √	0- √	0- √	0- √	0-√	
c	∞	5b	4a √	4a √	4a √	4a √	4a
d	∞	∞	∞	6c √	6c √	6c √	6c
e	∞	∞	∞	8c	8c √	8c √	8c
f	∞	∞	∞	∞	11d	9e√	9e

Implementation

Following enrichment, the four attributes are normalized individually across all nodes and mapped onto a range of 0-1 to streamline comparison and combination. The attributes for each node are summed to produce the edge weight. Following K-means clustering in the selection stage, the highest-scoring node from each cluster is chosen for the graph of candidate stations. A* search then computes the shortest path between stations. The resultant network is plotted taking the geographic coordinates of each station as its position, producing a geographic view of optimal stations and corridors alongside the cluster maps produced in the selection stage.

Results and Discussion

The framework was applied to three distinct-selected corridors to evaluate Hyperloop suitability under different economic, transportation, and regulatory conditions.

Case Studies

West Africa (Lagos–Accra): Identified as a freight-focused opportunity due to limited regional transport connectivity and growing trade demand. Existing ECOWAS cooperation and the planned Abidjan–Lagos Highway provide a foundation for future cross-border Hyperloop development.

Southeast Asia (Jakarta–Surabaya): Demonstrated strong passenger demand but highlighted the financial challenges of conventional high-speed rail. Hyperloop could provide a higher-capacity alternative while reducing pressure on existing transport infrastructure.

Middle East (Dubai–Riyadh): Emerged as the strongest candidate for Hyperloop implementation, supported by high passenger demand, strong economic connectivity, and ongoing regional infrastructure integration.

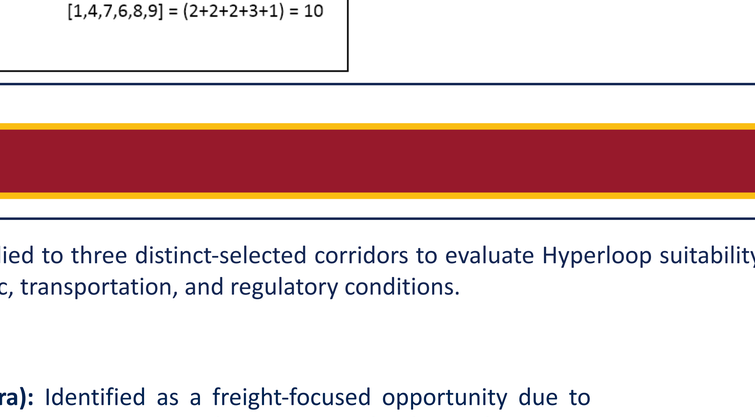
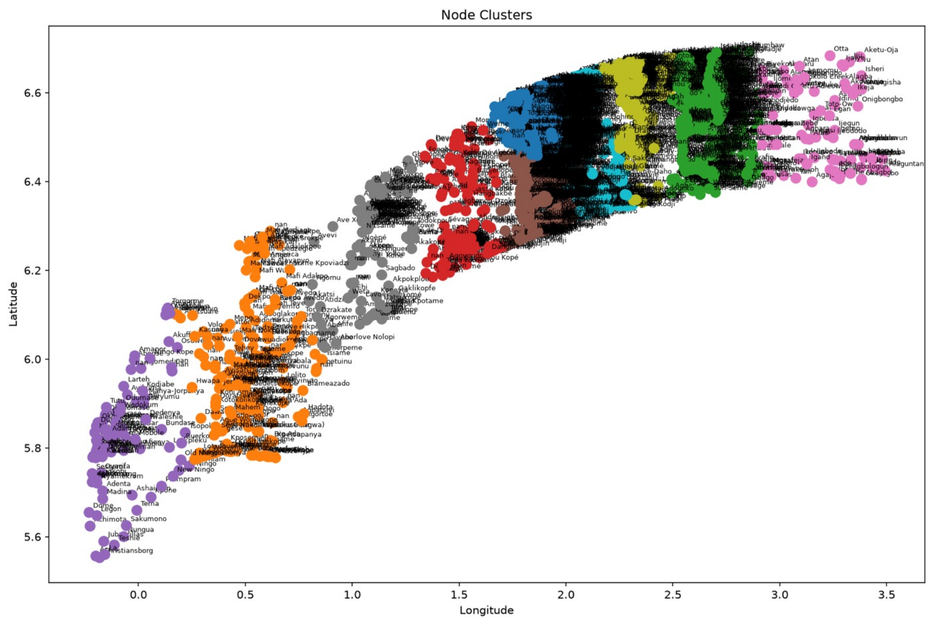


Figure 3: Focused map on the area of interest in West Africa. The Lagos-Accra corridor, as shown in the shaded region, is the geographic region of interest for the case study.



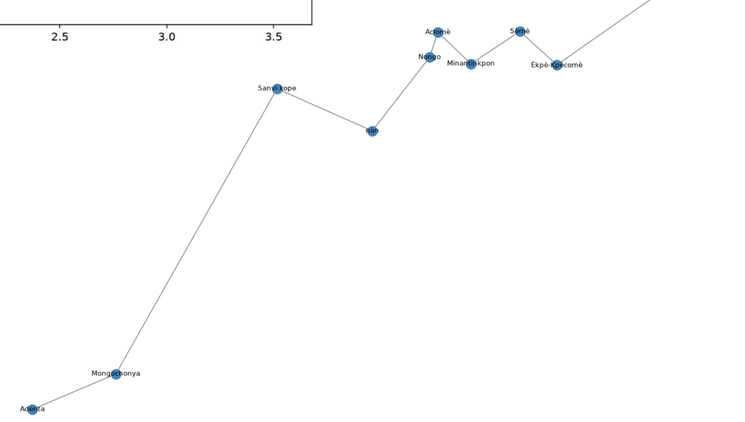


Figure 4 and 5: The node clusters output of the region of interest in West Africa, the Lagos-Accra corridor (top). The optimal route or node network of the Lagos-Accra corridor (bottom). This was determined through Dijkstra's algorithm as deployed by the adaptive python algorithm.

Overall Findings

The analysis shows that successful Hyperloop deployment depends on more than engineering feasibility. Corridors with high travel demand, strong economic activity, supportive policy environments, and strategic freight opportunities provide the greatest potential for long-term network value. The proposed framework offers a structured approach for prioritizing future Hyperloop investments based on measurable socio-economic and geographic factors.

Conclusion

This research developed an adaptive framework for Hyperloop corridor selection and passenger–freight allocation using geographic and socio-economic data. By combining terrain, connectivity, freight access, and population demand, the framework identifies viable station locations and evaluates network performance across diverse regions.

The case studies demonstrate that successful Hyperloop implementation depends on more than just engineering feasibility. Economic demand, regulatory readiness, and integration with existing transportation systems are equally important in determining long-term network value. The framework provides a flexible decision-deducing tool that can be adapted to different regions and strategic priorities

Recommendations

Future recommendations should focus on optimizing networks connections by incorporating route-level terrain and construction costs rather than relying on proximity alone. Develop dynamic scheduling models to simulate mixed passenger and freight operations across the network. In addition, economic modelling should be incorporated. This includes ticket pricing, operating costs, and financial feasibility. The dataset should also be expanded with higher-resolution terrain, environmental, demographic, and infrastructure data to improve corridor evaluation and station selection.